AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

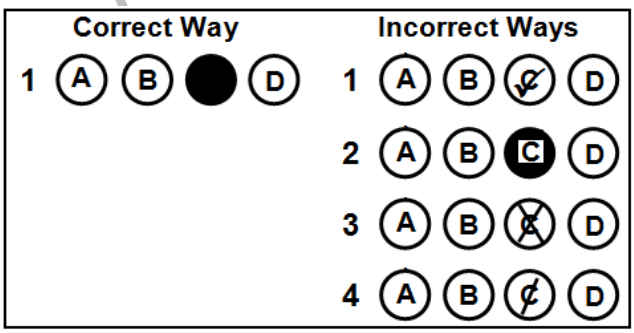
Biology Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. In the Northern parts of Pakistan, there are 45 lakes out of which 12 are enriched with trout fish.

   Which of the following biologists are most suited to study the physical and chemical properties of these habitats?

   A. Microbiologists
   B. Freshwater biologists
   C. Marine water biologists
   D. Environmental biologists

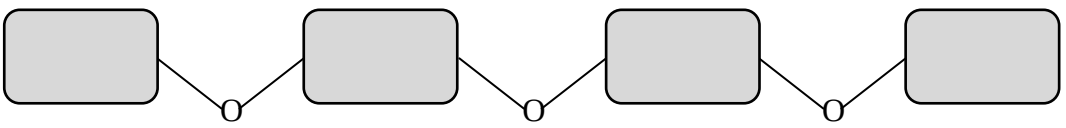
2. Which of the two levels of biological organisation are represented in the given diagrams?



I **II**

	I	II
A	Ecosystem	Population
B	Population	Community
C	Community	Population
D	Population	Ecosystem

3. How many molecules of water are required to hydrolyse the given part of a polysaccharide?



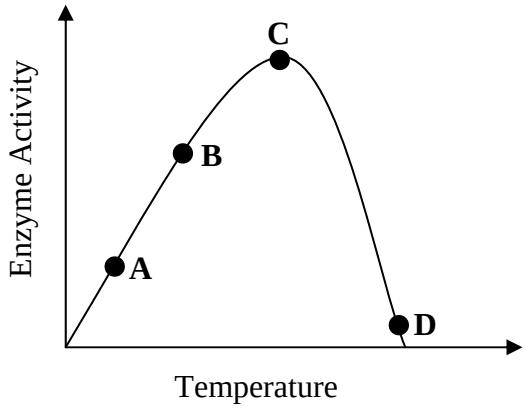
   A. 3
   B. 4
   C. 5
   D. 6

4. The property of water which results due to the formation of hydrogen bonds among its molecules is

   A. density.
   B. cohesion.
   C. ionisation.
   D. vaporisation.

5. The given graph shows the effect of temperature on enzyme activity.

   The point where the maximum kinetic energy has no effect on the structure of the enzyme is



6. Glyoxysomes are organelles found in plant cells which

   A. convert proteins into fats.
   B. are abundant in seedlings.
   C. are absent in lipid rich seeds.
   D. take part in anabolic reactions only.

7. Cytoskeleton is made up of microtubules, microfilaments and intermediate filaments.

   Which of the following is CORRECT about the structure, composition and function of microtubules?

	Structure	Composition	Function
A	Branched	Proteins	Maintenance of cell shape
B	Branched	Carbohydrates	Maintenance of cell shape
C	Unbranched	Proteins	Help in mitosis
D	Unbranched	Carbohydrates	Help in mitosis

8. During the lytic cycle, the function of lysozymes released by phage is to

   A. synthesise viral genome.
   B. dissolve bacterial cell wall.
   C. break down bacterial DNA.
   D. speed up the formation of proteins.

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9. When hand-written, the scientific name of the butterfly is written as

   A. Chaetodon ocellata.
   B. Chaetodon ocellata.
   C. Chaetodon Ocellata.
   D. Chaetodon Ocellata.

10. Which of the following is CORRECT about the components of the cell wall in Gram positive bacteria?

	Outer Membrane	**Peptidoglycan Layer**
A	Absent	Present
B	Present	Absent
C	Absent	Absent
D	Present	Present

11. Hydrogen peroxide inhibits the growth of microorganisms on non-living matter.

    Hydrogen peroxide acts as a/ an

    A. vaccine.
    B. antibiotic.
    C. antiseptic.
    D. disinfectant.

12. Penicillin restricts bacterial growth within the body of an infected person by inhibiting the formation of

    A. cell wall.
    B. cytoplasm.
    C. cell membrane.
    D. adenosine triphosphate.

13. The filaments of algae are made up of coenocytes which means that they

    A. do not have cross-walls.
    B. have leaf-like extensions.
    C. are composed of distinct cells.
    D. cannot be differentiated into roots and stems.

14. The organisms which contain naked cytoplasm are

    A. diatoms.
    B. red algae.
    C. water molds.
    D. myxomycots.

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15. The shells of foraminiferans contain pores through which cytoplasmic projections are extended.

These cytoplasmic projections help foraminiferans to

A. entangle the prey.
B. enhance water beauty.
C. form limestone deposits.
D. exchange genetic material.

16. In the sexual reproduction of fungal life cycle, the resultant hyphae of plasmogamy contains

A. two nuclei of same genetic type.
B. two nuclei of different genetic types.
C. one nucleus with diploid number of chromosomes.
D. one nucleus with haploid number of chromosomes.

17. The given statements represent different stages of reproduction in Zygomycotes.

I. Development of aerial hyphae with sporangium
II. Formation of gametangia
III. Mating of two different hyphae
IV. Formation of haploid spores
V. Development of zygospore from zygote

The CORRECT sequence of stages involved in the reproduction is

A. III, IV, V, II, I
B. III, IV, II, V, I
C. III, II, V, I, IV
D. III, II, IV, V, I

18. The cells of the embryo sac of an apple plant include

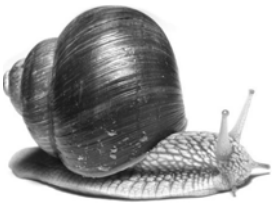
A. synergids and antipodal cells.
B. antipodal cells and tube cells.
C. synergids and generative cells.
D. tube cells and generative cells.

19. Which characteristic is TRUE about sporophyte and gametophyte in the life cycle of gymnosperms?

	Sporophyte	**Gametophyte**
A	Independent and dominant	Less conspicuous and dependent
B	Less conspicuous and dependent	Independent and dominant
C	Dependent and dominant	More conspicuous and independent
D	More conspicuous and independent	Dependent and dominant

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20.





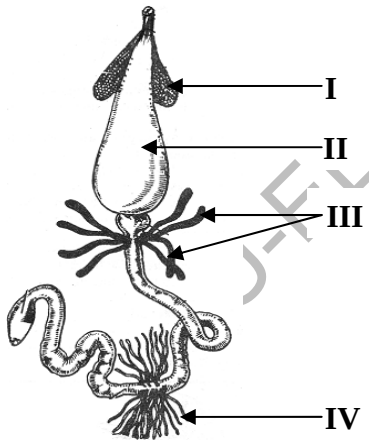
The organisms shown in the given diagrams are placed in

A. same phylum, same class.
B. same phylum, different class.
C. different phyla, same kingdom.
D. different phyla, different kingdom.

21. The CORRECT limiting factor(s) for the light independent reactions of photosynthesis is/ are

	Water	**Temperature**	**CO_2**
A	×	✓	✓
B	✓	✓	×
C	×	×	✓
D	✓	×	×

22. The given diagram represents the digestive system of a cockroach.

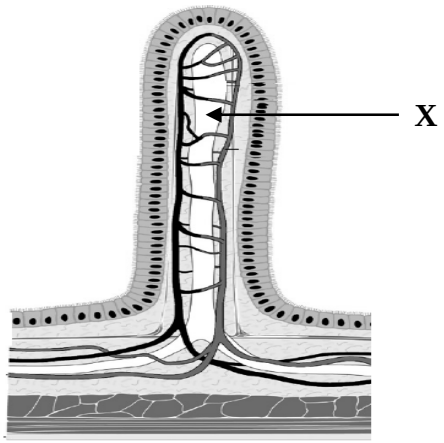


The structures which secrete digestive enzymes are

A. I and III
B. I and IV
C. II and III
D. II and IV

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23. The given diagram shows a villus of human intestine.



Which of the following conjugated molecules forms in the labelled part **X**?

A. Glycolipids
B. Lipoprotein
C. Glycoprotein
D. Nucleoprotein

24. What would be the site of the given reaction if it proceeds from right to left?

$$CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow H^+ + HCO^-_3$$

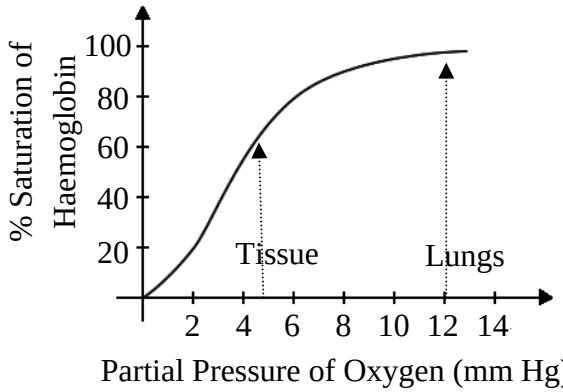
A. Lungs
B. Trachea
C. Body tissue
D. Blood plasma

25. The type of respiration which does NOT involve blood is

A. tracheal respiration in insects.
B. pulmonary respiration in frogs.
C. cutaneous respiration in earthworms.
D. bucco-pharyngeal respiration in frogs.

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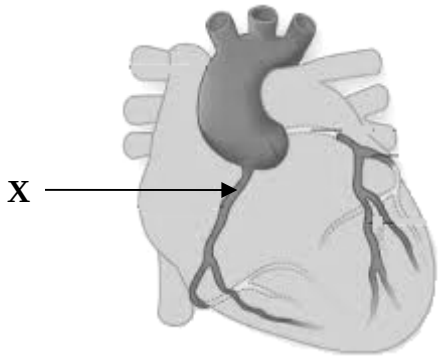
26. The given graph shows the haemoglobin saturation curve.



The relative oxygen concentration in body tissues and lungs as presented in the graph is

	Relative Oxygen Concentration in	
	Body Tissues	**Lungs**
A	Low	High
B	Low	Low
C	High	High
D	High	Low

27. The given diagram depicts the structure of a human heart.



The labelled part **X** in the diagram indicates

A. bundle of Hiss.
B. Purkinje fibres.
C. coronary artery.
D. chordea tendinae.

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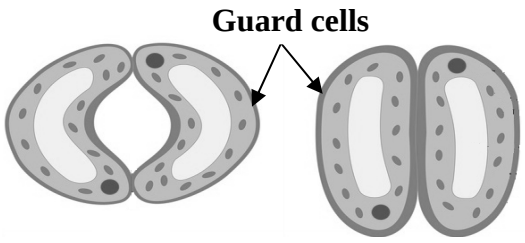
28. Given are the structures through which sucrose transports in a plant.

I. sieve tube cells.
II. companion cells.
III. bundle sheath cells.

The CORRECT pathway of sucrose transportation from source to phloem is

A. I → II → III
B. II → III → I
C. III → II → I
D. I → III → II

29. The given diagram represents guard cells of stomata in two different conditions.



Which of the following options is TRUE about the **conditions I** and **II** in guard cells?

	Condition I		**Condition II**	
	Concentration of K^+ ions	**Concentration of Water**	**Concentration of K^+ ions**	**Concentration of Water**
A	High	High	Low	Low
B	Low	Low	High	High
C	Low	High	High	Low
D	High	Low	Low	High

30. In the conduction of an impulse from sinoatrial node (S-A node) to atrioventricular node (A-V node), there is a delay of 0.15 seconds. This actually permits the

A. atrial systole to be completed.
B. atrial diastole to be completed.
C. ventricular systole to be completed.
D. ventricular diastole to be completed.

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